KYOUNGLIM CHO

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RESEARCH INTERESTS

Security of AI and machine learning systems; applied cryptography and key management; authentication and identity security; secure systems and infrastructure.

EDUCATION

Sejong Cyber University, Seoul, South Korea — Mar 2025 – Expected Aug 2027
Bachelor's Program, Department of Information Security (admitted as third-year transfer student)
Relevant Coursework: Introduction to Information Security; Unix and Linux Systems; Fundamentals of Cryptographic Algorithms; Network Security

Ewha Womans University, Seoul, South Korea — Mar 2015 – Feb 2022
B.A. in Philosophy; Double Major in Life Sciences

Additional Quantitative Coursework (Academic Credit Bank System, credit-bearing)
Linear Algebra; Probability and Statistics

PROFESSIONAL EXPERIENCE

Hana Bank, Seoul, South Korea — Apr 2023 – Present
Information Security Engineer (Assistant Manager), Digital Security Team, Information Security Department

Cryptographic Infrastructure & Key Management

- Operate the bank's enterprise data-encryption platform, covering database field-level encryption of regulated personal data and DMZ transport encryption, with HSM/KMS-based key management across production, development, and DR environments.
- Support application integrations using AES, ARIA, and SEED in CBC/GCM modes; installed and validated cryptographic modules on Linux, AIX, and Solaris; diagnosed C-module failures caused by file-permission and shared-memory behavior; wrote Java test code for developers integrating a legacy cryptographic library.
- Diagnosed intermittent key-generation failures in a newly deployed ATM encryption module; with the vendor engineer, traced the root cause to thread contention in a library function through kernel-level log analysis, then validated the fix with off-peak patching, real-time monitoring, and performance testing.
- Led the replacement of end-of-service encryption servers, including impact simulation, backup/recovery procedures with vendor engineers, and network-path coordination; completed the migration with no service interruption.

Authentication & Identity Security

- Operate the enterprise MFA service (directory-integrated PC agent with mobile OTP/push and facial authentication); troubleshoot DNS and application-process incidents.
- Extended MFA to overseas users to support the New York branch's NYDFS 23 NYCRR Part 500 compliance, and implemented SSO from PC logon to the group portal.
- Designed the DR architecture for the employee authentication app and coordinated system interfaces with senior systems engineers, delivering offline authentication during network outages and a DR server for service continuity.

Network & API Security

- Deployed proxy servers end to end (rack installation, cabling, IP and initial configuration); resolved an API outage by using Wireshark to identify blocked CORS preflight (HTTP OPTIONS) requests and reconfiguring the forwarding policy.
- Isolated access-gateway connectivity faults through DMZ packet captures and TCP handshake analysis; traced a post-replacement connectivity failure to a switch-side fault with network engineers.
- Assessed external OpenAPI integrations against authentication, endpoint access-control, and confidentiality requirements, and recommended remediation to partner organizations.

Governance & Leadership

- Prepare audit evidence for ISMS-P certification and financial-regulator examinations; negotiated maintenance contracts below the market benchmark rate and moved low-usage modules to per-call billing.
- Served as an IT Division Change Management Leader (2025); promoted collaboration-tool adoption to improve new-hire onboarding, knowledge sharing, and operational handoffs.

Sundosoft Co., Ltd., Seoul, South Korea Jan 2022 – Jun 2022
Data Analyst, Intelligent Information Business Division

- Developed an image-augmentation algorithm that automatically generated training images (e.g., plastic bottles) for a computer-vision waste-classification model, as part of a proposal for a government-funded AI project.
- Analyzed regional big data at the province and city/county/district levels using Python, SQL, and QGIS.

SELECTED MACHINE LEARNING PROJECTS

Team projects completed in the Big Data–based AI Solution Developer Program (Jun – Dec 2021); served as team leader for all three.

Korean Text Summarization for Literacy Training (ET5) | Team Leader Nov – Dec 2021

- Led a team building a web-based Korean summarization-practice program on ET5 (a Korean T5 model), using a publisher-provided book corpus and AI Hub open data.
- Personally ran fine-tuning experiments and hyperparameter tuning; achieved a best BERTScore F1 of 0.8391. Won the Top Excellence Award at a national AI service competition (see Honors).
  Code: github.com/Kyounglim-Joey/cakd3_3rd_Project

Mold Detection in Images (YOLOv5) | Team Leader 2021

- Led development of a mold-detection program with the YOLOv5 object-detection framework, including image annotation and preprocessing.
  Code: github.com/Kyounglim-Joey/cakd3_2nd_project-mold_detection

Predicting Declining-Spend Customers from Card Transaction Data | Team Leader 2021

- Led exploratory analysis and machine-learning prediction of customers with declining purchase activity, using Python and SQL.
  Code: github.com/Kyounglim-Joey/cakd3_1st_project-Data_exploration

PROFESSIONAL TRAINING

K-Shield Program, Incident Response Track, Korea Internet & Security Agency (KISA) Jun – Nov 2025
Web/mobile penetration testing with attack scenarios; web application vulnerability analysis; development of a vulnerability-assessment tool for critical information infrastructure; integrated wired/wireless penetration-testing lab build; generative AI for incident response; live-fire cyber exercise.

Big Data–based AI Solution Developer Program Jun – Dec 2021
Machine learning and deep learning in Python, SQL, and statistics for data analysis; three team projects (above).

HONORS & AWARDS

Grand Prize (1st place), CoP Category, Sangsang Ground Innovation Competition, Hana Bank 2024
Proposed a personalized financial-challenge platform that classifies users by spending patterns and recommends challenges with rewards.

First Place (Top Excellence Award), ONE-TEAM Payment Idea Competition, Hana Bank 2024
Proposed a salary-management consulting and gamification program to grow payment-product use among early-career customers.

Grand Prize (1st place), CoP Category, Sangsang Ground Innovation Competition, Hana Bank 2023
Proposed a digital sales-onboarding system enhancement, later implemented by the company's development team.

Top Excellence Award, 2021 AI Service Competition Using Korean Book Corpus, Woongjin Booxen Dec 2021

CERTIFICATIONS

Engineer Information Security (Korean national technical qualification)
ADsP – Advanced Data Analytics Semi-Professional, Korea Data Agency Nov 2024
Engineer Big Data Analysis (Korean national qualification) Dec 2021

TECHNICAL SKILLS & LANGUAGES

Security: HSM, KMS and key lifecycle management, AES/ARIA/SEED (CBC, GCM), MFA/SSO, API security assessment, ISMS-P
Systems & Networking: Linux, AIX, Solaris, TCP/IP, DNS, proxy and access gateways, Wireshark packet analysis
Programming & Data: Python, SQL (Oracle), Java, C, HTML, QGIS
Machine Learning: Transformer-based NLP (T5/ET5), object detection (YOLOv5), image data augmentation, BERTScore evaluation
Languages: Korean (native); English (OPIc Intermediate High, 2025); Chinese (HSK Level 4, 2025)